Project Report Format

1. INTRODUCTION

1.1 Project Overview

The **Software Installation Request System** is developed using the **ServiceNow** platform to automate the software request process within an organization. Employees can request licensed software through the **Service Catalog**, while managers and IT support teams handle approvals and installation through automated workflows. The system improves request tracking, reduces manual effort, and ensures faster software delivery.

1.2 Purpose

The purpose of this project is to provide a centralized and automated platform for software installation requests. It eliminates manual email-based approvals, improves transparency, reduces processing time, and ensures secure handling of software requests using ServiceNow workflows, catalog items, business rules, and catalog tasks.

2. IDEATION PHASE

2.1 Problem Statement

Employees often experience delays when requesting licensed software due to manual approval processes and lack of request tracking. This project automates the entire request lifecycle using ServiceNow, ensuring faster approvals, task assignment, and software installation.

2.2 Empathy Map Canvas

The project focuses on understanding employee needs while requesting software. Employees expect a quick and transparent process, managers require approval visibility, and IT teams need automated task assignment. The solution addresses these expectations through automation and real-time request tracking.

2.3 Brainstorming

Several solutions such as email approvals, manual ticket creation, spreadsheets, and automated workflows were discussed. The team selected a ServiceNow-based Service Catalog solution because it provides automation, approval workflows, notifications, task creation, and scalability.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. Employee opens Service Portal.
2. Searches for **Software Installation Request**.
3. Enters software details.
4. Submits the request.
5. Manager receives approval request.
6. Software Support Team receives installation task.
7. Software is installed.
8. Request is closed.

3.2 Solution Requirement

Functional Requirements

- Employee submits software request.
- Catalog variables collect software information.
- Manager approval workflow.
- Software Support task creation.
- Email notifications.
- Request tracking.
- Incident creation when license is unavailable.

Non-Functional Requirements

- Secure access.
- High availability.
- Fast response.
- Easy maintenance.
- Scalability.
- User-friendly interface.

3.3 Data Flow Diagram

The DFD illustrates how employee requests flow through the Service Catalog, approval workflow, business rules, and fulfillment tasks. Data is stored in **sc_request**, **sc_req_item**, and **sc_task** tables while notifications keep users informed.

3.4 Technology Stack

Layer	Technology
Platform	ServiceNow
Service Catalog	Catalog Items

Layer	Technology
Workflow	Workflow Editor / Flow Designer
Business Logic	Business Rules
Client Logic	UI Policies
Notifications	Email Notifications
Database	ServiceNow Tables
Tables	sc_request, sc_req_item, sc_task, incident

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The proposed solution automates software request management by replacing manual processes with ServiceNow Service Catalog workflows. Automated approvals, task assignments, and notifications improve request processing while ensuring transparency and compliance.

4.2 Proposed Solution

The solution enables employees to submit licensed software requests through the Service Portal. ServiceNow workflows automatically manage approvals, create fulfillment tasks, and notify users. Business Rules generate incidents when software licenses are unavailable, ensuring efficient issue management.

4.3 Solution Architecture

The architecture consists of:

- Employee
- Service Portal
- Service Catalog
- Catalog Variables
- Workflow
- Business Rule
- Manager Approval
- Software Support Team
- Catalog Task
- Incident
- ServiceNow Database
- Email Notification

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was developed using Agile methodology.

Sprint 1

- Requirement Analysis
- Service Catalog Creation
- Variables Configuration
- UI Policies

Sprint 2

- Workflow Creation
- Business Rules
- Incident Automation
- Testing

Sprint 3

- Service Portal Testing
- Documentation
- Update Set Migration
- Final Demonstration

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The project was tested for:

- Catalog Item Creation
- Variable Validation
- Approval Workflow
- Task Generation
- Incident Creation
- Email Notifications
- Service Portal Functionality
- Update Set Migration

Performance testing confirmed that requests were processed successfully without errors and completed within expected response times.

7. RESULTS

7.1 Output Screenshots

The project successfully demonstrated:

- Service Catalog Item
- Software Request Form
- Workflow Execution
- Approval Records
- Catalog Task Creation
- Incident Creation
- Service Portal Submission
- Request Status Tracking
- Email Notifications
- Update Set Export & Import

8. ADVANTAGES & DISADVANTAGES

Advantages

- Fully automated workflow.
- Faster approval process.
- Reduced manual effort.
- Real-time tracking.
- Better transparency.
- Easy maintenance.
- Secure role-based access.
- Scalable architecture.

Disadvantages

- Requires ServiceNow platform.
- Initial workflow configuration takes time.
- Licensed software depends on available licenses.
- Requires administrator permissions for customization.

9. CONCLUSION

The **Software Installation Request System** successfully automates the complete software request lifecycle using ServiceNow. Employees can submit requests through the Service Catalog, managers approve requests electronically, and IT teams receive automated fulfillment tasks. The project improves efficiency, reduces delays, enhances visibility, and provides a scalable solution for enterprise software request management.

10. FUTURE SCOPE

Future enhancements include:

- AI-based software recommendation.
- Integration with Software Asset Management (SAM).
- Microsoft Teams approval integration.
- Mobile application support.
- Dashboard analytics.
- Automatic software deployment using Integration Hub.
- License availability prediction.
- SLA monitoring and reporting.

11. APPENDIX

Source Code

- Service Catalog Item
- Catalog Variables
- UI Policies
- Business Rules
- Workflow
- Flow Designer
- Client Scripts

ServiceNow Tables

- sc_request
- sc_req_item
- sc_task
- incident

Project Resources

- Update Set XML
- Project Screenshots
- Service Portal Screenshots
- Workflow Screenshots
- Testing Report

Project Demo

- Service Portal Demonstration
- Workflow Execution
- Incident Creation
- Catalog Task Generation
- Update Set Migration

This structure is well suited for a college project report and aligns with your ServiceNow implementation.